

Felix Hoang

xuanphuc.algv@gmail.com | linkedin.com/in/phuchoang2603 | github.com/phuchoang2603 | phuchoang.sbs

Education

University of South Florida

Bachelor of Science in Computer Science, GPA: 3.6

Tampa, FL

Expected May 2028

Experience

Research Assistant

Sep 2025 - Present

Secure & Intelligent Systems Lab

- Automated the end-to-end research pipeline by engineering a data service with **mcp-[arxiv-server](#)**, converting raw Arxiv PDFs into structured Markdown for standardized documentation.
- Orchestrated an **n8n** workflow to programmatically inject datasets and benchmarks as YAML metadata, enabling granular content-based search across a library of **200+** papers.
- Streamlined knowledge distribution by deploying a research portal via **Hugo** and **GitHub Actions**, reducing the publication cycle from several hours to **under 2 minutes**.

Software Engineer

Jun 2024 – Nov 2024

Summit Education

- Built an e-waste e-commerce platform that automates device trade-ins, grading, and resale flows, integrating secure payment routing and partner fulfillment tracking.
- Developed high-performance RESTful APIs in **Go**, cutting end-to-end latency by **30%** through goroutine-based concurrency and channel-driven request batching while supporting **500+** concurrent storefront interactions.
- Optimized **PostgreSQL** schemas and queries, raising throughput **40%** for read-heavy marketplace operations and enabling predictable scaling for seasonal spikes.
- Implemented **Redis** caching for product catalogs and session state, halving database load on tier-1 endpoints and keeping UX response times under **200ms**.
- Architected an asynchronous **RabbitMQ** task system to process background jobs, improving reliability and freeing APIs to serve customer traffic without blocking on billing and fulfillment workflows.
- Containerized Go services with **Docker** and deployed to **Kubernetes**, accelerating deployment frequency while maintaining consistent observability across releases.

Projects

End-to-End MLOps Platform for Real-Time Fraud Detection

- Architected a **FastAPI** inference service on **GKE** handling **2,000+ QPS** with a **P99 latency of 50ms**, using **Terraform** for dynamic resource scaling.
- Implemented a **GitOps** workflow with **ArgoCD** and **Helm** to automate model hot-swapping, reducing deployment lead time to **under 15 minutes** with **zero downtime**.
- Deployed a **Prometheus/Grafana** observability stack with automated data-validation gates to detect model drift and minimize false positives in high-traffic production environments.

Secure & Automated RKE2 Kubernetes Bootstrap | *Terraform, Ansible, Vault, External Secrets*

- Automated High Availability (HA) RKE2 provisioning on **Proxmox VE**, slashing the bootstrap workflow from **4 hours** to **15 minutes** through reusable **Terraform** modules and **Ansible** playbooks.
- Triggered Terraform and Ansible from **GitHub Actions** using ephemeral self-hosted runners authenticated via **HashiCorp Vault OIDC**, letting workflows fetch credentials dynamically.
- Secured the cluster API surface and pod workloads with the **External Secrets Operator**, keeping every credential short-lived while avoiding static kubeconfigs.
- Unified Traefik ingress and Longhorn storage into a single declarative YAML pipeline, ensuring consistent configuration drift control across every cluster life cycle stage.

Technical Skills

DevOps & Infrastructure: Kubernetes, Docker, Terraform, Ansible, ArgoCD, Helm, GitHub Actions, Linux

Observability & Tools: Prometheus, Grafana, Loki, Tempo, Alloy, HashiCorp Vault, External Secrets Operator

Cloud & Virtualization: Google Cloud Platform (GCP), AWS, Proxmox VE, TrueNAS

Programming & Databases: Go (Golang), Python (FastAPI), Bash scripting, SQL, PostgreSQL (CloudNativePG)